

Chapter 11 — Active learning basics

Active learning attacks the labeling bottleneck by letting the model choose which unlabeled items humans should label

Learning objectives

- By the end of this part

The iterative feedback loop

- Active learning starts from a small labeled seed set
- The cycle repeats until performance or budget stops the loop
- The goal is better generalization with fewer total labels than random annotation

Query strategies overview

- A query strategy defines how the model ranks unlabeled items for human labeling
- Uncertainty sampling targets low-confidence predictions near decision boundaries
- Margin sampling ranks by the gap between the top two class probabilities
- Entropy sampling ranks by predictive entropy over the full class distribution
- Strategies can be combined or adapted to the task

Example 11.1 — Uncertainty Sampling for Cat-Dog Images

- Example 11.1 — hands-on module
- Example 11.1 shows uncertainty sampling on a binary vision task
- An image with predicted probabilities near 0.51 and 0.49 for cat versus dog is queued for
- Explore the chapter example module
- View files: ``modules/chapter11/example1/``

Example 11.1 — listing

```
"""Example 11.1 – uncertainty sampling queues near-tie binary
predictions."""

from __future__ import annotations

def uncertainty_score(p_positive: float) -> float:
    """Return distance from a 0.5 decision boundary (lower => more
    uncertain)."""
    return abs(p_positive - 0.5)

def main() -> None:
    """Rank unlabeled images by binary prediction uncertainty."""
    pool: list[tuple[str, float]] = [
        ("img_clear_dog.jpg", 0.92),
        ("img_clear_cat.jpg", 0.11),
        ("img_clear_person.jpg", 0.51)
```

Example 11.2 — Margin Sampling in Multi-Class Tasks

- Example 11.2 — hands-on module
- Example 11.2 refines the idea for multi-class settings
- Signaling weak class separation that a new label can sharpen
- This differs from absolute least-confidence alone by focusing on the competing pair of
- View files: ``modules/chapter11/example2/``

Example 11.2 — listing

```
"""Example 11.2 – margin sampling prioritizes the smallest top-two
gap."""

from __future__ import annotations

def margin(probs: dict[str, float]) -> float:
    """Return the gap between the top two class probabilities."""
    ordered = sorted(probs.values(), reverse=True)
    return ordered[0] - ordered[1]

def main() -> None:
    """Rank multi-class predictions by margin (smaller margin => query
    first)."""
    pool: list[tuple[str, dict[str, float]]] = [
        ("doc_clear", {"sports": 0.80, "politics": 0.12, "biz": 0.08})
    ]
```


Example 11.3 — High-Entropy Multi-Class Prediction

- Example 11.3 — hands-on module
- Example 11.3 illustrates entropy sampling
- If the model predicts class A at 0.4 and classes B and C at 0.3 each
- Flat distributions across many classes are informative even when no single probability is
- View files: ``modules/chapter11/example3/``

Example 11.3 — listing

```
"""Example 11.3 – entropy sampling selects the flattest predictive
distribution."""

from __future__ import annotations

import math

def entropy(probs: list[float]) -> float:
    """Return Shannon entropy of a discrete probability
    distribution."""
    return -sum(p * math.log2(p) for p in probs if p > 0)

def main() -> None:
    """Compare entropy for peaked vs flat multi-class predictions."""
    cases: list[tuple[str, list[float]]] = [
        ("peaked (A 0.99)", [0.99, 0.05, 0.05]),
```

Active-learning pipeline steps

- The pipeline is incremental
- The loop stops when the model meets a performance threshold or the labeling budget is
- Effort concentrates on informative cases rather than easy majority examples

Example 11.4 — Active Learning for MRI Tumor Detection

- Example 11.4 — hands-on module
- Example 11.4 applies the full loop to expert MRI labeling
- A tumor detector starts from a small labeled MRI seed set
- Explore the chapter example module
- View files: ``modules/chapter11/example4/``

Takeaways

- Active learning is an iterative loop that queries humans where the model is least sure
- Uncertainty, margin, and entropy sampling are complementary ways to rank informativeness
- Medical and vision examples show how seed training plus selective expert labels can reduce

Next

- Complete the quiz for this part
- The next part covers complementary techniques beyond basic uncertainty